

Module 05: Action in Robotics — Control Law Design

Learning Objectives

1. Define robotic **control action** as the process of selecting motor commands that minimize the gap between the estimated state and the desired state.
2. Analyze how classical control laws (PID, LQR, MPC) relate to Active Inference action selection — proprioceptive prediction error minimization.
3. Apply the **precision-compliance trade-off** to design controllers that balance rigid goal pursuit with adaptive compliance.

Introduction

Action in the control-estimation framework is the robot's output — the motor commands sent to actuators that change the physical state of the world. In Active Inference, action arises as the solution to a simple imperative: make the predicted proprioceptive state come true. If the generative model predicts that the robot's arm should be at position X, the motor system generates the commands needed to bring the actual arm position to X.

Key Concepts

1. PID Control as Prediction Error Minimization

The PID controller — the most widely deployed controller in industrial robotics — has a direct Active Inference interpretation:

- **Proportional (P)**: Acts on the current prediction error — the difference between desired and actual state
- **Integral (I)**: Acts on the accumulated prediction error — correcting for persistent systematic bias (steady-state error)
- **Derivative (D)**: Acts on the rate of change of prediction error — damping oscillations by anticipating where the error will be

In Active Inference terms, P is the prediction error, I corrects for unmodeled constant disturbances (bias in the generative model), and D is a first-order prediction of future prediction error. The PID gains are precision parameters — they determine how aggressively the controller responds to errors.

2. Linear-Quadratic Regulation (LQR) as Optimal Policy

LQR finds the control policy that minimizes a quadratic cost function (weighted sum of state deviation and control effort) for a linear system. In Active Inference terms:

- The state deviation cost corresponds to the **C vector** (preferred observations) — how much does the actual state deviate from the preferred state?
- The control effort cost corresponds to the **policy precision** — how costly is it to exert control?
- The LQR solution provides the optimal gain matrix — the precision-weighted mapping from state error to control action

LQR is Active Inference when the generative model is linear-Gaussian, the cost is quadratic, and the time horizon is infinite.

3. Model Predictive Control (MPC) as Planning

MPC looks ahead: at each timestep, it optimizes the control sequence over a finite horizon, then executes only the first step:

- The "model" in MPC is the generative model's B matrix — used to simulate future state trajectories
- The "predictive" aspect is the EFE evaluation — predicting the consequences of each candidate action sequence
- The "control" is the selection and execution of the EFE-minimizing first action

MPC naturally handles constraints (actuator limits, obstacle avoidance, joint limits) and multi-objective trade-offs. It is the engineering implementation of Active Inference planning for continuous control systems.

4. Impedance Control: Precision Meets Physics

Impedance control specifies the desired relationship between position error and force:

- **High impedance** (stiff): The robot resists perturbations and tracks the desired trajectory precisely → high precision on proprioceptive predictions
- **Low impedance** (compliant): The robot yields to external forces and adapts to contact → low precision on proprioceptive predictions, high precision on force/tactile predictions
- **Variable impedance**: The controller modulates stiffness based on the task phase — stiff during free-space movement, compliant during contact

This is directly the Active Inference precision modulation: the same mathematical structure that explains how the brain adjusts the gain of reflex arcs.

Applications

- **Collaborative robot (cobot) safety**: A cobot working alongside humans uses impedance control — maintaining low impedance (compliance) during close human interaction so that accidental contact produces gentle yielding rather than injurious force. The compliance is precision modulation on proprioceptive prediction errors.
- **Precision machining**: A CNC milling robot uses high-impedance control (high precision on trajectory prediction errors) during cutting passes, achieving micrometer-level position accuracy. Between passes, it switches to lower impedance for tool changes and workpiece repositioning.

Conclusion

Control action in robotics spans PID (reactive prediction error minimization), LQR (optimal steady-state policy), MPC (predictive planning), and impedance control (precision-compliance modulation). All are implementations of Active Inference action — making proprioceptive predictions come true. The next module examines how robotic learning adapts these controllers to new tasks.